THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. At a party, 15 people each hug every other person exactly once. Number of hugs: 1

(a) 210 **(b)** 120 **(c)** 105 **(d)** 112

2. A clock shows 2 o'clock. After 3 hours, the smaller angle between the clock hands is: 1

(a) 60 **(b)** 90 **(c)** 150 **(d)** 210

3. How many steps does the Kaprekar routine take to reach 6174 from 3141? 1

(a) 7 **(b)** 6 **(c)** 4 **(d)** 5

4. The total of the table, after removing the largest value, is: 1

Item	Number
Banana	75
Mango	25
Orange	15
Apple	55
Grapes	45

(a) 145 **(b)** 140 **(c)** 200 **(d)** 215

5. The smallest 4-digit number divisible by 13 is: 1

(a) 1001 **(b)** 1002 **(c)** 1014 **(d)** 988

6. A rectangle's length is 2 times its breadth and its perimeter is 54 cm. Its area (cm^2) is: 1

(a) 171 **(b)** 153 **(c)** 162 **(d)** 180

7. $\frac{5}{6} + \frac{3}{8} - \frac{1}{4} =$ 1

(a) $1\frac{11}{24}$ **(b)** $\frac{17}{24}$ **(c)** $\frac{23}{24}$ **(d)** $1\frac{1}{6}$

8. Two concentric circles have radii 4 cm and 9 cm. A point is 0 cm from the common centre. It lies in the ring between the circles only if its distance is more than 4 cm and less than 9 cm. How many of the distances 3, 4, 6, 9, 8, 12 cm give a point in the ring? 1

(a) 4 **(b)** 6 **(c)** 3 **(d)** 2

9. A regular polygon has an angle of rotation 40° and its perimeter is 45 cm. The side length is (cm): 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

10. -14 - (-9) + (22) - (-5) = 1

(a) 20 **(b)** 24 **(c)** 32 **(d)** 22

11. If $1 + 2 + 3 + \cdots + n = 703$, then n is: 1

(a) 38 **(b)** 36 **(c)** 37 **(d)** 19

12. A wheel turns 3 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 1260 **(b)** 1350 **(c)** 1080 **(d)** 1170

13. The hailstone (Collatz) sequence of 25 has how many terms (including 25 and 1)? 1

(a) 23 **(b)** 24 **(c)** 22 **(d)** 25

14. One symbol of a pictograph stands for 10. Class A has 12 symbols and its total is 3 times Class B. How many symbols does Class B have? 1

(a) 36 **(b)** 9 **(c)** 5 **(d)** 4

15. The number of composite numbers from 1 to 30 (both included) is: 1

(a) 21 **(b)** 18 **(c)** 20 **(d)** 19

16. A square and an equilateral triangle have equal perimeters. If the square's side is 12 cm, the triangle's side (cm) is: 1

(a) 20 **(b)** 16 **(c)** 28 **(d)** 12

17. $\frac{3}{4}$ of a number minus $\frac{1}{2}$ of it is 15. The number is: 1

(a) 70 **(b)** 12 **(c)** 30 **(d)** 60

18. A rectangle's diagonal is 12 cm. The perimeter of triangle AOB where OA = OB = 6 cm and AB = 8 cm is (cm): 1

(a) 26 **(b)** 21 **(c)** 20 **(d)** 19

19. After a rotation of 135° a regular 8-gon coincides with itself. This rotation is 3 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 75 **(b)** 55 **(c)** 45 **(d)** 35

20. The sum of the integers from -5 to 9 (both included) is: 1

(a) 30 **(b)** 40 **(c)** 31 **(d)** 29

21. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 400 **(b)** 890 **(c)** 800 **(d)** 779

22. A spinner starts pointing North and turns clockwise 270° then anticlockwise 90°. The final angle clockwise from North is: 1

(a) 270 **(b)** 180 **(c)** 190 **(d)** 170

23. The number of 4-digit palindromes with first digit 2 is: 1

(a) 90 **(b)** 100 **(c)** 10 **(d)** 9

24. A bar graph has scale 1 cm = 12 units. The heights of two bars differ by 7 cm. The values differ by: 1

(a) 19 **(b)** 84 **(c)** 96 **(d)** 72

25. The smallest number divisible by each of 6, 8 and 12 is: 1

(a) 25 **(b)** 24 **(c)** 30 **(d)** 23

26. A rectangular field 24 m by 10 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 96 **(b)** 144 **(c)** 120 **(d)** 130

27. A drum is $\frac{3}{4}$ full. 20 litres are removed and it becomes $\frac{1}{4}$ full. The capacity (litres) is: 1

(a) 39 **(b)** 40 **(c)** 50 **(d)** 41

28. A circle of radius 3 cm is drawn. A square of side equal to the diameter is drawn around it. The area of the square (cm^2) is: 1

(a) 39 **(b)** 36 **(c)** 33 **(d)** 42

29. The number of letters among A, H, B, I, C with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

30. On a number line, the distance between -14 and 9 is: 1

(a) 26 **(b)** 23 **(c)** 22 **(d)** 24

31. The difference between the 20th and the 19th square numbers is: 1

(a) 41 **(b)** 39 **(c)** 37 **(d)** 20

32. The smaller angle between the hands of a clock at 1 o'clock is 30°. The smaller angle between the hands at 11 o'clock is: 1

(a) 30 **(b)** 330 **(c)** 60 **(d)** 150

33. The sum of 10 consecutive counting numbers starting from 12 is: 1

(a) 165 **(b)** 120 **(c)** 175 **(d)** 220

34. A school bus carries 30 children on trip 1, 45 on trip 2 and 60 on trip 3. A pictograph uses one symbol for 15 children. The number of symbols needed for all three trips is: 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

35. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 26 **(b)** 25 **(c)** 24 **(d)** 23

36. A picture 30 cm by 20 cm is mounted on a frame of width 3 cm all around. The area of the frame alone (cm^2) is: 1

(a) 336 **(b)** 333 **(c)** 339 **(d)** 348

37. A rod 6 m long is cut into pieces of $\frac{3}{4}$ m each. The number of pieces is: 1

(a) 7 **(b)** 9 **(c)** 8 **(d)** 10

38. Two circles of equal radius 4 cm touch externally. The distance from the outer end of one to the outer end of the other along the line of centres (cm): 1

(a) 16 **(b)** 12 **(c)** 20 **(d)** 24

39. The sum of the angles of rotation of a square and a regular hexagon (in degrees) is: 1

(a) 140 **(b)** 180 **(c)** 150 **(d)** 160

40. Two integers add to -5 and one of them is 12. The other is: 1

(a) -7 **(b)** -17 **(c)** -15 **(d)** -19

41. **Assertion (A):** In the sequence $3, 7, 11, 15, \ldots$ the next term is 19. 1
Reason (R): Each term of this sequence is obtained by adding 4 to the term before it.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** An angle of 135° is an obtuse angle. 1
Reason (R): An obtuse angle is more than 90° and less than 180°.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** The largest number that can be made from the digits 3, 0, 7, 5 (each used once) is 7530. 1
Reason (R): To make the largest number, arrange the digits in descending order.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** In a pictograph where one symbol stands for 10 children, 3 and a half symbols show 35 children. 1
Reason (R): A half symbol stands for half of the value of a whole symbol.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The number 1 is neither prime nor composite. 1
Reason (R): The number 1 has only one factor, namely itself.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The perimeter of a rectangle 12 m long and 5 m wide is 34 m. 1
Reason (R): Perimeter of a rectangle = 2 x (length + breadth).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** $\frac{3}{4}$ and $\frac{6}{8}$ are equivalent fractions. 1
Reason (R): Multiplying the numerator and denominator by the same non-zero number gives an equivalent fraction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The diameter is the longest chord of a circle. 1
Reason (R): A diameter passes through the centre of the circle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A square has 4 lines of symmetry. 1
Reason (R): A regular polygon with n sides has n lines of symmetry.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** $-7 < -3$. 1
Reason (R): On a number line, a number lying to the left of another number is smaller.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**